

1. Basic Math / Number Theory

- ✓ Check if a Number Is Positive, Negative, or Zero
- ✓ Find the Largest of Three Numbers
- ✓ Check if a Year Is a Leap Year
- ✓ Find if a Triangle Is Valid Based on Angles or Sides
- ✓ Perform Unit Conversion (e.g., Cm to M, Kg to G)
- ✓ Find the Sum of the First N Natural Numbers
- ✓ Check if a Number Is Prime
- ✓ Check if a Number Is a Palindrome
- ✓ Check if a Number Is an Armstrong Number
- ✓ Calculate the Gcd and Lcm of Two Numbers
- ✓ Generate All Prime Numbers Between 1 and N
- ✓ Find the Factorial of a Number
- ✓ Find the Sum of Natural Numbers
- ✓ Find the Sum of Digits of any Number
- ☐ **Find Integers A, B Such That $N = A^3 \times B$, Where B Is Cube-Free**

2. Conditionals / Characters / Switch

- ✓ Count number of digits in a number
- ✓ Add each digit of a number in a ArrayList/array
- ✓ Given a point (x,y) - write a program to find out its quadrant or axis or at the origin.
- ✓ Check if a Character Is a Vowel or Consonant
- ✓ Check for Uppercase, Lowercase, or Special Character
- ✓ Reverse a String
- ✓ Check Whether a String Is Palindrome or Not
- ✓ Find Duplicate Characters in a String

3. Arrays – Basics & Elementary Operations

- ✓ Print Fibonacci Sequence and Nth Term of Fibonacci Series
- ✓ Print Numbers From 1 to N and N to 1
- ✓ Check if an Array Is Sorted
- ✓ Find the Second Largest Element in an Array
- ✓ Reverse an Array

- ☒ Remove All the Duplicate Elements From an Array
- ☒ Move All the Zeros to the End
- ☒ Find Leaders in an Array
- ☒ Find the Frequency in a Sorted and Unsorted Array
- ☐ Find the Maximum Difference (✗)
- ☐ Left Rotate an Array by D Places (✗)

4. Arrays – Subarrays, Sliding Window, Prefix Sum

- ☐ Find the Maximum Consecutive 1's in a Binary Array
- ☐ Find the Majority Element in an Array
- ☐ Find the Maximum Subarray Sum (Kadane)
- ☐ Sliding window (subarray sum)
- ☐ Even-odd subarray
- ☐ Prefix sum
- ☐ Find the Maximum or Minimum Sum of a Subarray of Size K
- ☐ Find a Subarray With a Given Sum Using the Sliding Window Technique
- ☐ Find the Maximum Length Even-Odd Subarray
- ☐ Find the Sum of Any Subarray Using the Prefix Sum Array
- ☐ Find the Equilibrium Point

5. Arrays – Two Pointers / Hashing / Misc Advanced

- ☐ Sort an Array Which Consists of Only 0, 1, and 2 Without Using Sorting
- ☐ Find the Union and Intersection of Two Sorted Arrays
- ☐ Find All Pairs in an Array Whose Sum Is Equal to a Given Number
- ☐ Find All Elements That Appear More Than N/k Times in an Array
- ☐ Find Common Elements in 3 Sorted Arrays
- ☐ Solve the Chocolate Distribution Problem
- ☐ Rearrange Positive and Negative Integers Alternately in an Array
- ☐ Create Another Array With the Product of All Elements Except the i th Element
- ☐ Find a Peak Element in an Array
- ☐ Find if There Is a Pair With Sum X in a Sorted Array
- ☐ Find a Triplet in a Sorted Array
- ☐ Solve the Trapping Rain Water Problem
- ☐ Find the Best Time to Buy and Sell Stock

6. Bit Manipulation

- ☐ Check if the Kth Bit Is Set
- ☐ Count Set Bits
- ☐ Check if a Number Is a Power of Two
- ☐ Find the Only Odd Occurring Number
- ☐ Find Two Odd Appearing Numbers
- ☐ Find the Missing Number in an Array
- ☐ Count the Number of Bits to Be Flipped to Convert a to B
- ☐ Find the Xor of Numbers From L to R

7. Matrices (2D Arrays)

- ☐ Print a Matrix in a Snake Pattern
- ☐ Print Boundary Elements of a Matrix
- ☐ Print the Transpose of a Matrix
- ☐ Rotate a Matrix by 90 Degrees
- ☐ Perform Spiral Traversal of a Matrix

8. Searching Sorting

- ☐ Perform Binary Search

9. Recursion

Basic Recursion Problems

- ☐ Find the Power of Two Using Recursion
- ☐ Count the Digits in a Number Using Recursion
- ☐ Reverse a String Using Recursion
- ☐ Check if a String Is a Palindrome Using Recursion
- ☐ Factorial of a number
- ☐ Fibonacci series (nth term / full series)
- ☐ Sum of first n natural numbers
- ☐ Sum of digits of a number
- ☐ Product of digits of a number
- ☐ Find GCD (HCF) using recursion

- ☐ Binary search using recursion
- ☐ Print numbers from n to 1 and 1 to n
- ☐ Calculate a^b using optimized recursion (fast exponentiation)

String-Based Recursion

- ☐ Remove all occurrences of a character from a string
- ☐ Replace all occurrences of a substring
- ☐ Remove consecutive duplicate characters
- ☐ Move all 'x' to the end of the string
- ☐ Generate all substrings of a string
- ☐ Generate all permutations of a string
- ☐ Print all subsequences of a string : **Output:** "", "a", "b", "c", "ab", "ac", "bc", "abc"

Array-Based Recursion

- ☐ Check if an array is sorted
- ☐ Find maximum/minimum element in array
- ☐ Linear search using recursion
- ☐ Find all indices of a given element
- ☐ Reverse an array recursively
- ☐ Generate balanced parentheses
- ☐ Merge sort

Backtracking-Type (Very Important)

- ☐ Generate all subsets (power set)
- ☐ Combination sum problem
- ☐ N-Queens problem
- ☐ Rat in a maze problem
- ☐ Sudoku solver
- ☐ Word search problem

Advanced / Interview Level

- ☐ K-th symbol in grammar
- ☐ Letter combinations of a phone number
- ☐ Recursive stack problems (sort stack, reverse stack)
- ☐ Quick sort
- ☐ Tower of Hanoi

☐ Josephus problem